Free balancing equations worksheet worksheet 18

Maths | 7th grade | Algebra

1. If 3(x - 4) = 2x + 5, what is the value of x?

A) 17
B) 19
C) 13
D) 15

2. Solve for y: 5y + 3 = 2y + 18.

A) 5
B) 3
C) 6
D) 7

3. What is the solution of 8 - 2z = 4?

A) -2
B) 2
C) 4
D) -4

4. The sum of three consecutive numbers is 72. What is the smallest of these numbers?

A) 23
B) 22
C) 24
D) 25

5. If 7m - 4 = 3m + 12, what is the value of m?

A) 4
B) 5
C) 6
D) 3

6. If 3a + 12 = 8a - 3, find a.

A) 3
B) 2
C) 5
D) 4

7. For which value of c is the equation 4(c - 1) = 2c + 6 true?

A) 5
B) 4
C) 3
D) 6

8. If 9 - 2b = b + 15, find the value of b.

A) -2
B) -3

C) 8
D) 6

9. The difference between three times a number and 7 is 17. Find the number.

A) 8
B) 9
C) 7
D) 6

10. When 6 is subtracted from twice a number, the result is 10. What is the number?

A) 7
B) 8
C) 6
D) 5

Answer Key

1. D
Distribute and simplify: 3x - 12 = 2x + 5. Then, 3x - 2x = 5 + 12. So, x = 17.

2. A
Subtract 2y from both sides: 3y + 3 = 18. Subtract 3: 3y = 15. Divide by 3: y = 5.

3. A
Subtract 8 from both sides to get -2z = -4. Then divide by -2: z = 2.

4. B
Let the numbers be x, x+1, x+2. Then x + (x+1) + (x+2) = 72. Solve 3x + 3 = 72. x = 23.

5. C
Subtract 3m from both sides to get 4m - 4 = 12. Add 4: 4m = 16. Divide by 4: m = 4.

6. D
Subtract 3a from both sides: 12 = 5a - 3. Add 3: 15 = 5a. Divide by 5: a = 3.

7. C
Distribute: 4c - 4 = 2c + 6. Subtract 2c: 2c - 4 = 6. Add 4: 2c = 10. Divide by 2: c = 5.

8. D
Add 2b to both sides: 9 = 3b + 15. Subtract 15: -6 = 3b. Divide by 3: b = -2.

9. A
Let the number be x. Solve 3x - 7 = 17. Add 7: 3x = 24. Divide by 3: x = 8.

10. D
Let the number be y. Solve 2y - 6 = 10. Add 6: 2y = 16. Divide by 2: y = 8.